homage to us!”—and concluding that our pretensions are amusing, our aspirations pathetic, that this must be the planet of the idiots.

But such a judgment is too harsh. We did the best we could. There was an unlucky coincidence between everyday appearances and our secret hopes. We tend not to be especially critical when presented with evidence that seems to confirm our prejudices. And there was little countervailing evidence.

In muted counterpoint, a few dissenting voices, counseling humility and perspective, could be heard down through the centuries. At the dawn of science, the atomist philosophers of ancient Greece and Rome—those who first suggested that matter is made of atoms—Democritus, Epicurus, and their followers (and Lucretius, the first popularizer of science) scandalously proposed many worlds and many alien life forms, all made of the same kinds of atoms as we. They offered for our consideration infinities in space and time. But in the prevailing canons of the West, secular and sacerdotal, pagan and Christian, atomist ideas were reviled. Instead, the heavens were not at all like our world. They were unalterable and “perfect.” The Earth was mutable and “corrupt.” The Roman statesman and philosopher Cicero summarized the common view: “In the heavens … there is nothing of chance or hazard, no error, no frustration, but absolute order, accuracy, calculation and regularity.”

Philosophy and religion cautioned that the gods (or God) were far more powerful than we, jealous of their prerogatives and quick to mete out justice for insufferable arrogance. At the same time, these disciplines had not a clue that their own teaching of how the Universe is ordered was a conceit and a delusion.

Philosophy and religion presented mere opinion—opinion that might be overturned by observation and experiment—as certainty. This worried them not at all. That some of their deeply held beliefs might turn out to be mistakes was a possibility hardly considered. Doctrinal humility was to be practiced by others. Their own teachings were inerrant and infallible. In truth, they had better reason to be humble than they knew.